Fintech Case Study

Indian Fintech Ecosystem

UPI, AI, and the Future of Digital Payments

Presented by Swetha Battula
Course: Calibro AI Academy
Date: March 14, 2026

Made with GAMMA

What is Fintech?

Digital Platform

Banking, payments, lending through mobile apps

Financial Access

Reduces dependency on traditional banking

Market Growth

$50B in 2024 → $150B by 2025

Fintech turns your phone into a bank branch

Made with GAMMA

From Cash to Tap

India's Payment Journey

1. **1990s–2000s**
   Cash-dominated economy, limited card penetration
2. **2010–2015**
   Mobile wallets rise with smartphone adoption
3. **2016**
   Demonetization accelerates digital payments
4. **2016–Present**
   UPI becomes world's largest real-time payment system

In a decade, India leaped from cash to global fintech leadership

Made with GAMMA

UPI – The Backbone of Digital India

14B

Monthly Transactions

As of early 2024

2016

Launched by NPCI

Unified Payments Interface

2s

Instant Transfer

Bank-to-bank in seconds

UPI made digital payments simpler than handing over cash

Made with GAMMA

Rohan's ₹30 Journey

Inside a UPI Transaction

Scan QR → App Processing → Bank Auth → NPCI Routing → Vendor Bank

Six steps, two seconds, zero friction

Made with GAMMA

AI – The Invisible Guardian

Protecting Digital Payments

Fraud Detection

Real-time suspicious pattern identification

Risk Scoring

Instant transaction risk assessment

System Monitoring

Predictive infrastructure protection

Personalization

Tailored financial products

AI works silently so your payment works instantly

Made with GAMMA

Growing Pains

Challenges at Scale

Scalability

500+ transactions per second stress infrastructure

Fraud Risk

₹1,087 crore lost to UPI fraud in FY 2023-24

Downtime

Server outages during festivals disrupt millions

Interoperability

Ensuring seamless cross-platform experiences

Scale brings opportunity—and vulnerability

Made with GAMMA

UPI Sentinel

An AI-Powered Stability Layer

Spike Prediction

Forecasts transaction surges

Fraud Detection

Identifies anomalies

Load Balancing

Prevents server overloads

Real-Time Alerts

Notifies before breaches

Predict, protect, stabilize—payments that never fail

Made with GAMMA